

Malaviya National Institute of Technology Jaipur

Department of Physics

End term Examination (Odd Semester 2024-25)

Course Name: ITR and Cosmology

Course Code: PHT-415

Class : B.Tech (Open Elective)

Max.Marks:50

Time:2.5 hrs

Date:4/12/2024

Answer all Questions

1. What is big bang theory? write the supporting evidences of Big bang theory? Explain the short comings of Big bang model? [10M]
2. Describe the Michaelson-Morley experiment with neat diagram and explain the results of it? [10M]
3. Explain the space time diagram and Geometrical interpretation of Lorentz transformations? [10M]
4. What is red shift and blue shift? Explain the Gravitational lensing? [5M]
5. A rod is moving with speed $\frac{c}{2}$ with respect to ground such that velocity of rod is parallel to its length. An observer is also running in the same direction with speed $\frac{c}{3}$ with respect to ground. If rest length of rod is L_0 . Find the apparent length of the rod as measured by the observer? [5M]
6. A Particle of rest mass m_0 and kinetic energy $2m_0c^2$ collides with a stationary particle of rest mass $2m_0$ and after collision they coalesce together. Find the rest mass of the coalesced particle? [5M]
7. What is the percent error in the classical expression of kinetic energy when a body is moving with a speed $0.5c$? [5M]

End

$$\frac{M_0}{\sqrt{1 - \left(\frac{v'}{c}\right)^2}}$$

$$m' = m \left(\sqrt{1 - \left(\frac{v'}{c}\right)^2} \right)$$

$$\frac{mv}{2m_0} = \frac{m'v'}{2m_0}$$

$$2\sqrt{2}m_0c = m'v'$$



MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY, JAIPUR

Department of Electronics & Communication Engineering
End-Term Examination, B.Tech 7th Semester
Subject: Satellite Communication & Radar Engineering (ECT670)

Time Allowed – 2:30 hrs.

MM-50

Important Instructions:

- Write your Roll No. on the space provided in the question paper.
- This question paper contains 8 questions. All questions are compulsory.
- Figures to the right indicate maximum marks.

- Q: 1)** Explain Satellite launch in detail. List different launch vehicle selection factors. 5
- Q: 2)** Explain Satellite Antenna? Why Cassegrain antenna popular for large earth stations? 5
- Q: 3)** Explain briefly : 10
- Geosynchronous Satellites
 - Prograde and Retrograde
 - Sun Transit Outage
 - Transponders
 - VSAT
- Q: 4)** What is the operation of RADAR? Classify the types of RADAR systems? 5
- Q: 5)** What is TTC&M? Explain the working? 5
- Q: 6)** Explain briefly : 10
- RADAR Range
 - Pulse Repetition Frequency
 - Maximum Unambiguous Range
 - Dwell Time
 - Hits Per Scan
- Q: 7)** Calculate the look angles to a geostationary satellite. The details of the earth station site and sub satellite point are as follows: 5
- Earth station:**
- latitude: 28.06280° N
longitude: 80.62311° W
radius: 6370 km
- Sub Satellite Point:**
- latitude: 49.5432° N
longitude: 48.2967° W
radius: 38000 km
- Q: 8)** Explain Payload of Satellite? 5

hits $\uparrow$ as $f \uparrow$ $\frac{10}{4}$

$\frac{1}{180}^\circ$

Now
SOAR $\rightarrow$ re
sound $\rightarrow$ the

TTC $\rightarrow$ km



$$\cos \delta = (0.88243)(0.4887)(0.8450)$$

1.234567890

Malaviya National Institute of Technology, Jaipur

Department of Electronics & Communication

B. Tech - 7 semester/M.Tech (VLSI/Embedded) - Course : VLSI Signal Processing Architecture (21ECT833)

Marks 50

END TERM EXAM

Time: 2 hr 30 mins

Assume data if finds anything missing

Q-1 Draw the space-time representation and the systolic architecture of the design W1 with [8 Marks]

$$\mathbf{d} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{p}^T = \begin{bmatrix} 0 & 1 \end{bmatrix}, \text{ and } \mathbf{s}^T = \begin{bmatrix} 2 & 1 \end{bmatrix}.$$

Q-2 Draw the DFGs for the direct form and data broadcast forms of the FIR filter $y(n) = a*x(n) + b*x(n-2) + c*x(n-5)$ and unfold these two DFGs using unfolding factor $J=3$. [8 Marks]

Q-3 The Life Time Table for one design is given as below. Draw the forward backward allocation Table and folded architecture for $N=9$. [8 Marks]

Sample	Tinput	Life Period
a	0	0 to 4
b	1	1 to 7
c	2	2 to 10
d	3	3 to 5
e	4	4 to 8
f	5	5 to 11
g	6	6 to 6
h	7	7 to 9
i	8	8 to 12

Q-4 Draw architectures of linear carry select adder and carry skip adder compare the performance [8 Marks]

Q-5 Consider the 6th-order orthogonal filter structure shown in Fig.1. All operations in this structure are CORDIC (coordinate rotation digital computer) rotation operations, which are orthogonal. Assume that each CORDIC rotation operation requires T nsec. What is the critical path of this filter? Retime the filter structure to reduce the critical path. Show all the cutest locations used for re-timing explicitly. [8 Marks]

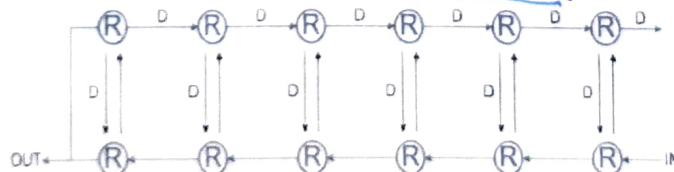


Fig.1 6th-order orthogonal filter structure

Q-6 (a) How pipeline and parallel processing help to reduce power consumption? Explain with examples. [5 Marks]

(b) Consider the DFG shown in Fig.1 [5 Marks]



Fig.2 Original DFG

- Find out the loop bonds of the DFG.
- What is the iteration bound of the DFG.
- Calculate the critical path and computational time for the same.

Total Marks: 50

Time: 2.5 h

Date: 05/12/2024

Answer all the Questions

1. (a) Discuss addition and condensation polymers with examples? What are isotactic, syndiotactic and atactic polymers? (3+3=6 marks)
- (b) An imaginary polymer material consists of 100 macromolecules. The molecules are composed of a single monomer type with a distribution in mass of the chain as given in the table below. Calculate the number average molar mass, mass average molar mass, and the polydispersity index of the polymer (3+3+4=10 marks).

No. of macromolecules	Mass of the chain
8	200
22	750
45	820
21	680
4	230

2. Discuss the behavior of polymers in solution w.r.t to their concentration? Calculate the average volume occupied by a polymer chain that has 90000 identical segments, each of which is spaced 1.5 nm apart. (3+3= 6 Marks).
 3. What are thermotropic and lyotropic liquid crystals? Explain the Smectic liquid crystalline phase? (2+2= 4 marks).
 4. Why are proteins considered as a class of soft material? Discuss the primary, secondary and tertiary structure of proteins. (2+4 =6 Marks).
 5. What are the two basic elements to the DNA molecule? Explain the formation of the double helix structure of the DNA? (2+2 =4 marks).
 6. Explain the basic working principle of a Transmission electron microscope (TEM)? How does it differ from an AFM? Discuss, with an example, how negative staining helps in imaging soft biological materials using a TEM. (2+2+2=6 marks)
 7. Calculate the fractional contribution of Debye, Keesom and London dispersion forces for aniline from the given data at 25 °C. (6 marks)
- $$\mu \text{ (Debye)} = 1.35 \quad \frac{ax10^{30}}{4\pi\epsilon_0} \text{ (m}^3\text{)} = 10.3 \quad \beta \times 10^{77} \text{ (Jm}^6\text{)} = 5.09$$
8. How does protein provide stabilization to milk? What is an emulsion? (3 marks)
 9. Explain the molecular organization of the lipid bilayer and discuss its function? (3 marks)
 10. (a) The Brownian force acting on a colloidal particle of size 250 nm at 25 °C is -----
 (b) ----- is an example of an acidic amino acid.

$$\mu \text{ (Debye)} = 1.35$$

$$\frac{qx10^{30}}{4\pi\epsilon_0} (\text{m}^3) = 10.3$$

$$\beta \times 10^{77} \text{ (Jm}^6\text{)} = 5.09$$

$$R_{K\Delta} = \frac{\frac{1}{2} m \omega^2}{T}$$

Er Ma

$$\frac{m \cdot a \cdot d}{\text{Kelvin}} = k_B \quad F = \frac{k_B \times T}{d}$$



Malaviya National Institute of Technology Jaipur
Department of Electronics and Communication Engineering
End-Term exam
VII Semester B.Tech. (ECE)
ECT620 Microelectronic Devices and Circuits

Max marks: 50

7th December, 2024

Instructions:

1. Interpret any bad data (if any).
2. Attempt all questions.
3. All questions carry equal marks

Student ID: _____

4/ 1. (a) Explain the working of bipolar junction devices with their current characteristics curve in each mode.

(b) What are the channel length modulation effects, and how do they impact the resistance of MOSFET?

2. (a) (i) Consider a silicon sample doped with $N_D = 2 \times 10^{15}/\text{cm}^3$ donor atoms. Assume that the intrinsic carrier concentration $n_i = 1.5 \times 10^{10}/\text{cm}^3$. If the sample is additionally doped with $N_A = 1 \times 10^{18}/\text{cm}^3$ acceptor atoms, calculate the approximate number of electrons/ cm^3 in the sample at $T = 300 \text{ K}$.

(ii) Write the equations of Fermi level for n-type and p-type semiconductor.

(b) What are the long channel effects? Explain with device modeling?

3. (a) What are the short channel effects, how these effects influence the working of device?

How to avoid the short channel effects in a device? *Dibl, V_{sat} , hot carrier CLM*

(b) What are the parasitic effects in MOSFET?

th. V.

4. (a) Explain the working operations of HEMTs and MESFETs devices.

(b) Explain the working of P-MOSFET in accumulation mode with energy band diagram. What are the noise margin in CMOS inverter and draw the CMOS based NAND & XNOR logic gate.

5. (a) Explain the Gummel-Poon BJT model?

(b) What is the difference between VMOS and DMOS structures?

$$I_C = I_S \left(e^{\frac{V_{be}}{V_T}} - 1 \right) \left(1 + \frac{V_{ce}}{V_0} \right)$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$(\overline{A}\overline{B}) \cdot (\overline{A}B) \\ (\overline{A}B)(AB)$$

7/1

$$(A+B)(A+B) \\ A\overline{B} + \overline{A}B$$



MALVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Advanced Semiconductor Devices (21ECT541)

Mid-Semester Examination: 27th September 2024

Time Allowed – 1:30 Hrs

MM-30

Notes:

- This question paper contains 6 questions.
- Marks are indicated against each question.
- Question no. 01 is compulsory.
- Attempt any four questions between Q. No. 02, 03, 04, 05 & 06.

Course Outcomes:

At the end of the course the student will be able to:

- CO1 Analyse MOSFET functionalities and physics
- CO2 Understand Short Channel Effects (SCEs)
- CO3 Design Scaled MOSFETs.
- CO4 MOSFET Modeling for scaled devices.
- CO5 Familiarisation with state of art semiconductor devices.

S. No.	Question Description	Marks	Mapped Course Outcome
1.	Differentiate between following (Briefly): a) Direct Band Gap Semiconductors v/s Indirect Band Gap Semiconductors b) Intrinsic v/s Doped Semiconductors c) HEMTs v/s MESFETs d) MOSFET v/s BJT e) Long Channel v/s Short Channel Transistors f) MOSFET as a Voltage Variable Resistor v/s MOSFET as an Amplifier g) Rectifying v/s Ohmic contact	2*7=14 Marks	CO1, CO2, CO3 & CO4
2.	Explain the structure and fundamental operation of MESFET. How this is different from MOSFET?	04 Marks	CO1, CO2 & CO5
3.	Explain CMOS fabrication steps and briefly define Latch-up problem in CMOS logic design.	04 Marks	CO2, CO3 & CO4

<u>4.</u>	Explain the fundamental operation of HEMT and its applications.	04 Marks	CO1, CO2, & CO5
<u>5.</u>	Explain C-V Characteristics of MOSFET and its variation with respect to frequency.	04 Marks	CO1 & CO2
<u>6.</u>	Explain various applications of Nanotechnology in the field of Electronics.	04 Marks	CO5

Malaviya National Institute of Technology, Jaipur

Materials Research Centre

Mid Semester Examination, B. Tech 4th Year (Odd Semester)

Paper Name: Introduction to Soft Materials (21MST805) (Open elective)

Total Marks: 30

Time: 90 minutes

Date: 26/09/2024

Answer all the Questions.

1. What do you mean by the terms dispersion, suspension, and solution? (3 marks).
2. Explain the process of nanoprecipitation with suitable example? Why is the size of colloids measured using dynamic light scattering (DLS) greater than that measured using transmission electron microscopy (TEM)? (2+2=4).
3. What is Brownian motion in colloids? What are the conditions for choosing perfect particles for studying Brownian motion? Using a suitable example, explain the concept of electrical double layer (1+3+3 = 7 marks).
4. Calculate the fractional contribution of Debye, Keesom and London dispersion forces for *t*-butanol from the given data at 25 °C. (6 marks)
 μ (Debye) = 1.67 $\frac{\alpha \times 10^{-30}}{4\pi\epsilon}$ (m³) = 9.46 $\beta \times 10^{-77}$ (Jm⁶) = 5.46
5. Although ΔH for micelle formation is positive, micelles do form. Explain? What are the factors that affect CMC. (2+3 = 5 marks)
6. How does the CMC affect the turbidity and molar conductivity of surfactants? Show that the CPP of surfactants in spherical micelles is equal to or less than 1/3. (3+2=5 marks).

3.33 x 10³⁰

$\frac{k_B T}{a}$

50.97 x 100
~~2.58 x 10³⁰~~
50.97 x 10



$\frac{V_h}{V_c}$
 $\frac{V_h}{V_c}$
 $\frac{V_h}{V_c}$

Malaviya National Institute of Technology Jaipur
Department of Physics
 Mid term Examination (Odd Semester 2024-25)

Course Name: ITR and Cosmology
 Course Code: PHT-415
 Class : B.Tech (Open Elective)

Max.Marks:30
 Time:1.5 hrs
 Date:25/09/2024

Answer all Questions

1. Derive an expression for the relativistic addition of velocities for the Components of u_x , u_y and u_z ? [6M]
2. Derive the Lorentz transformations? When the Lorentz transformations reduces to Galilean transformations? [6M]
3. Describe the Michaelson-Morley experiment with neat diagram and explain the results of it? [6M]
4. A Spaceship is moving at $0.6c$ relative to Earth. A Second spaceship is moving at $0.7c$ relative to the first spaceship. What is the velocity of the second spaceship as observed from Earth? [6M]
5. A rod of proper length 20 meters is moving at $0.95c$ relative to a stationary observer. The rod is aligned at an angle of 30° with respect to its direction of motion (not parallel to the motion). What is the observed length of the rod as measured by the stationary observer [6M]

End

Handwritten notes and diagrams:

Diagram 1: Two spaceships moving relative to Earth (S). One moves at $0.6c$, the other at $0.7c$.

Diagram 2: A rod of length $20m$ moving at $0.95c$ relative to a stationary observer (S). The rod is at an angle of 30° to its direction of motion.

Equations and formulas:

- $u_x = 0.6c$
- $x_1 = \gamma(x'_1 + vt'_1)$
- $x_2 = \gamma(x'_2 + vt'_2)$
- $L = \gamma(L_0)$
- $L > L_0$
- $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$
- $v \cos 30^\circ$
- $v \sin 30^\circ$
- $L = \gamma L_0$ (moving)
- $\cos 30^\circ = \frac{\sqrt{3}}{2}$
- $L_1 = \frac{L_0}{\sqrt{1 - \frac{v^2}{c^2}}}$



Roll No. _____

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY, JAIPUR

Department of Electronics & Communication Engineering

Mid-Term Examination, September 2024

B.Tech 7th Semester

Subject: Satellite Communication & Radar Engineering (ECT670)

Time Allowed – 1:30 hrs.

MM-30

Important Instructions:

- Write your Roll No. on the space provided in the question paper.
- This question paper contains 5 questions. All questions are compulsory.
- Figures to the right indicate maximum marks.

- Q: 1** How satellite orbit are defined? What is the difference between geosynchronous and a geostationary satellite? **6**
- Q: 2** List out the frequency bands used for satellite services. **6**
- Q: 3** State and explain Kepler's laws. Write down the Kepler's orbital elements? **6**
- Q: 4** What are the look angles? Explain how look angles are determined using sub-satellite points? Derive the necessary expressions for look angles. **6**
- Q: 5** A Satellite in an elliptical orbit around the earth has an apogee 41174 km and a perigee of 550 km. What is the orbital period of this satellite? Give answer in hours. (Note: Radius of the earth is 6,378.137 km Kepler's Constant = $3.9860004418 \times 10^5 \text{ km}^3/\text{s}^2$) **6**

Handwritten solution for Q: 5:

$$\mu = G M_E$$
$$G M_E = \frac{4\pi^2 a^3}{T^2}$$
$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$
$$T = \frac{2\pi (41174 + 550)^{3/2}}{\sqrt{3.9860004418 \times 10^5}}$$
$$T = 20.27 \text{ sec.}$$



Malaviya National Institute of Technology Jaipur
Department of Electronics and Communication Engineering

Mid-Term exam
VII Semester B.Tech.

ECT620 Microelectronic Devices and Circuits

September 23, 2024

Max marks: 30

Instructions:

1. Interpret any bad data (if any).
2. Attempt all questions.

Student ID:

1. (a) Explain how the band formation takes place with an orbital diagram also elaborate how it happens in direct and indirect band semiconductor materials? (3)
(b) What is the fermi-energy level and how does the position of fermi level varies in intrinsic and extrinsic semiconductors with mathematical formulae. (3)
2. (a) What are high-frequency devices? Discuss the properties of VMOS and DMOS devices with their structure. (3)
(b) What are the parasitic effects in MOS devices? Draw the energy band diagram of the metal- semiconductor junction of n-type semiconductor. (3)
3. (a) What are interface effects and subthreshold region in MOS devices, and how do the interface trap effects influence the electrical parameters of the FET device? (3)
(b) What are the MOS Capacitances? Draw the capacitance vs gate voltage characteristics of a simple MOS capacitor at low frequency. (3)
4. (a) Silicon is doped with boron to a concentration of 2×10^{17} atoms/cm³. Assume the intrinsic carrier concentration of silicon to be 1.5×10^{10} /cm³ and the value of kT/q to be 25 mV at 300 K. Compared to undoped silicon, the fermi level of doped silicon goes ___ by ___ eV. (3)
(b) Explain the working of N-MOSFET in depletion mode and CMOS Inverter with voltage transfer characteristic curve. (3)
5. What are the HEMTs and MESFETs devices, explain their working operation. (6)

$$f(r) = \ln\left(\frac{N_A N_D}{n_i^2}\right) \left(\frac{kT}{2q}\right)$$

25mV = $\ln$
Linear Saturation
inversion



Time Allowed – 2:30 Hrs

MM-50

Notes:

- This question paper contains 10 questions.
- Marks are indicated against each question.

Course Outcomes:

At the end of the course the student will be able to:

- CO1 Analyse MOSFET functionalities and physics
- CO2 Understand Short Channel Effects (SCEs)
- CO3 Design Scaled MOSFETs.
- CO4 MOSFET Modeling for scaled devices.
- CO5 Familiarisation with state of art semiconductor devices.

S. No.	Question Description	Marks	Mapped Course Outcome
1.	Explain the fundamental operation of Junctionless transistor and its applications.	05 Marks	CO3, CO4 & CO5
2.	Define all types of leakage problems in conventional MOSFETs.	05 Marks	CO1, CO2, & CO3
3.	Describe electrical nature of MS contact with respect to type of semiconductor material and work function of the material.	05 Marks	CO2 & CO3
4.	Explain the fundamental operation of HEMT and it's applications.	05 Marks	CO1, CO2, & CO3
5.	Explain short channel effects in detail.	05 Marks	CO1 & CO2
6.	What is SOI technology and explain its advantages and limitations over conventional MOSFETs.	05 Marks	CO3, CO4 & CO5

DI/BK

7.	Explain various applications of Nanotechnology.	05 Marks	CO3
8.	Explain various structures of Tunnel FET and its fundamental operation.	05 Marks	CO3, CO4 & CO5
9.	Explain the fundamental operation of FinFET and its applications.	05 Marks	CO3, CO4 & CO5
10.	Define process corners and environmental variations for non-ideal transistors.	05 Marks	CO1, CO2, & CO3